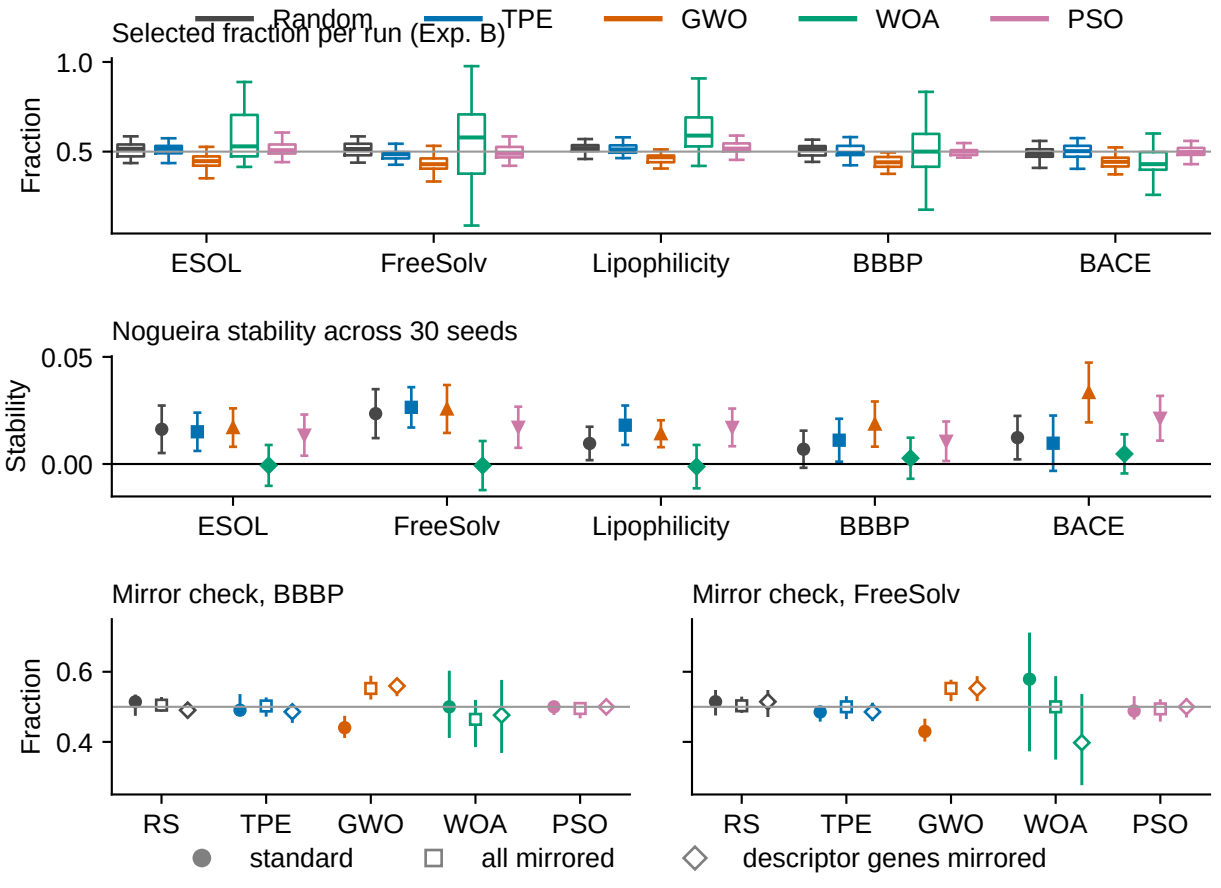
Random
TPE
GWO
WOA
PSO
Selected fraction per run (Exp. B)
Fraction
1.0
0.5
ESOL
FreeSolv
Lipophilicity
BBBP
BACE
Nogueira stability across 30 seeds
Stability
0.05
0.00
ESOL
FreeSolv
Lipophilicity
BBBP
BACE
Mirror check, BBBP
Mirror check, FreeSolv
Fraction
0.6
0.4
RS
TPE
GWO
WOA
PSO
standard
all mirrored
descriptor genes mirrored